

Chan Nirak Choun

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EDUCATION

University of California, Davis

Master of Science in Computer Science (Thesis Track)

Davis, CA

Sep. 2026 – Expected Jun. 2028

Fort Hays State University

Bachelor of Science in Computer Science

Hays, KS

Feb. 2022 – Dec. 2025

American University of Phnom Penh

Bachelor of Science in Information Technology Management

Phnom Penh, Cambodia

Feb. 2022 – Dec. 2025

Dual Degree Cumulative GPA: 3.78 / 4.0

PROJECTS

KernelForge — GPU Kernel Performance Analysis | *CUDA C++, Triton, cuBLAS, Slurm*

May. 2026

- Optimized FP32 SGEMM through six CUDA kernel versions (coalescing, shared-memory tiling, 2D register blocking, vectorized loads) on an NVIDIA RTX PRO 6000 Blackwell, reaching 72% of cuBLAS.
- Wrote memory-bound kernels (reduction, softmax, RMSNorm) in CUDA and Triton at 93–100% of measured DRAM bandwidth; the reduction matched NVIDIA CUB within 0.6%.
- Compared autotuned Triton with hand-written CUDA through PTX/SASS analysis: Triton reached 93% of cuBLAS but fell to 81% on shapes not divisible by 16, due to narrower loads and register spills.

TensorForge — MLIR Tensor Compiler for CPU and GPU | *MLIR/LLVM, C++, TableGen, PTX*

Jul. 2026

- Built an MLIR compiler for a custom tensor dialect (TableGen ops, verifiers, folds) lowering through Linalg to x86 and to NVIDIA GPUs via NVVM/PTX, validated with lit/FileCheck tests and FP64 references.
- Generated fused matmul+bias+ReLU GPU kernels with Transform-dialect tiling, shared-memory promotion, and vectorization, matching Triton and beating cuBLAS plus a separate epilogue by 9% on irregular shapes.
- Evaluated an analytical tile-selection cost model against autotuning over 840 runs on two GPU architectures; model picks reached 81% (Blackwell) and 78% (Ampere) of the autotuned best.

ArchForge — gem5 Hardware-Software Co-Design Study | *gem5, C, Python, Slurm*

Sep. 2026

- Built a gem5 x86 simulation framework (parameterized core and memory configs, 11 self-checking workloads, Slurm sweep pipeline) and ran 495 validated simulations.
- Ran a 190-configuration tile-size $\times$ cache co-design study: tiling gave $3.2\times$ vs. $2.5\times$ from a $4\times$ larger L2, joint tuning added only 6%, and tiling cut sensitivity to cache capacity from $2.5\times$ to $1.23\times$.
- Isolated microarchitectural mechanisms (ROB size vs. memory-level parallelism, prefetching, data layout, false sharing) and root-caused a gem5 artifact: MinorCPU never consults its branch predictor on x86.

EXPERIENCE

Undergraduate Research Assistant

Sep. 2024 – May 2025

Fort Hays State University

Hays, KS

- Evaluated 16 regression models for student-performance prediction with preprocessing and hyperparameter tuning; Support Vector Regression performed best ($R^2 = 0.713$).

IoT Developer Intern

May 2024 – Aug. 2024

FuseGap

Remote

- Developed C++ and Python computer-vision and monitoring components for resource-constrained in-vehicle cameras.
- Designed a C++ module for real-time health and battery monitoring of in-vehicle cameras.

TECHNICAL SKILLS

Languages: C++, C, CUDA C++, Python, x86-64 assembly (reading)

GPU Systems: CUDA, Triton, cuBLAS, PTX/SASS, Nsight Systems, roofline analysis

Compilers: MLIR, LLVM, TableGen, Linalg, NVVM, lit/FileCheck

Architecture: gem5, cache and memory-system modeling, microbenchmark design

Tools: Linux, Slurm, Git, CMake, SCons, GCC, Clang, GDB, Bash

Python: PyTorch, NumPy, pandas, Matplotlib